

Achieving Oracle Rate for Large Covariance Matrix Estimation From Quadratic Measurements

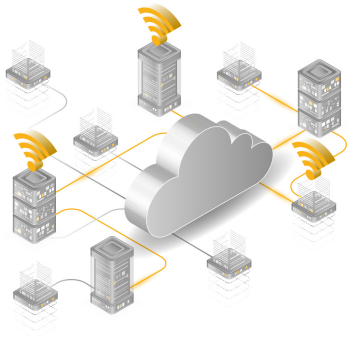
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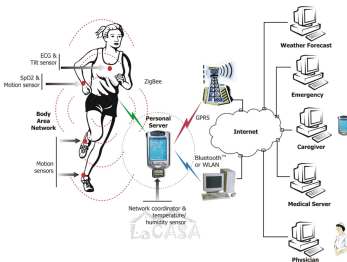


High-dimensional Streaming Data

Each time a snapshot of a data vector $\mathbf{x} \in \mathbb{R}^d$ is generated, with d large.



(a) Internet monitoring¹



(b) Wireless health monitoring²



(c) YouTube ranking³

¹<https://www.epitiro.com/>

²Jovanov, Emil, et al. "A wireless body area network of intelligent motion sensors for computer assisted physical rehabilitation." Journal of NeuroEngineering and rehabilitation 2.1 (2005): 6.

³<https://incartmarketing.com/youtube-ranking-how-to-get-more-views-on-youtube/>

Challenges in Modern Data Acquisition

Data generation at unprecedented rate: data samples are

- not observable due to privacy or security constraints;
- distributed at multiple locations;
- online generated on the fly and can only be accessed once.

Limited processing power at sensor platforms:

- **time-sensitive:** impossible to obtain a complete snapshot of the system;
- **storage-limited:** cannot store the whole data set;
- **power-hungry:** minimize the number of observations.

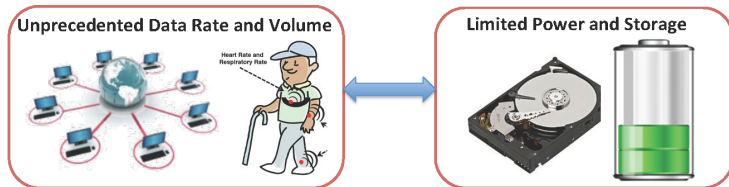


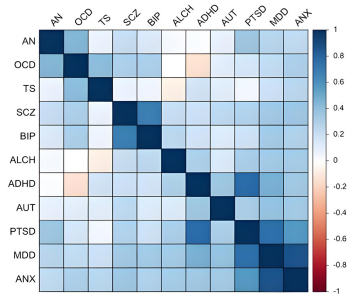
Figure 1: Mismatch streaming⁴

⁴<https://users.ece.cmu.edu/yuejie/GeometricConstraints.html>

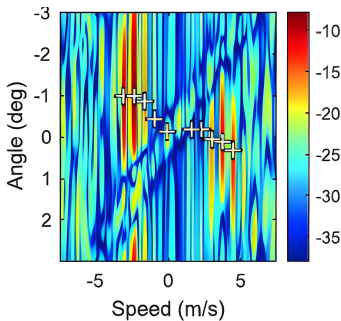
Covariance Sketching

Key Observation: The covariance structure can be recovered without measuring the whole data stream.

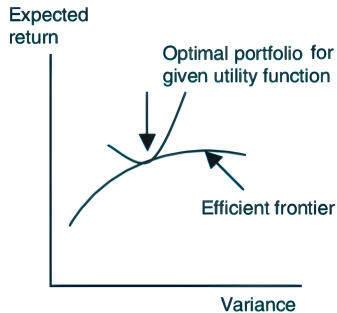
Applications of covariance sketching:



(a) Genetic Biology



(b) Radar System



(c) Portfolio Optimization

Quadratic Sketching for Covariance Estimation

Consider a data stream possibly distributedly observed at m sensors:



Quadratic Sketching: For each sensor $i = 1, \dots, m$:

- randomly select a sketching vector $\mathbf{a}_i \in \mathbb{R}^d$ with i.i.d. sub-Gaussian entries;
- Sketching n independent observations $\{\mathbf{x}_t\}_{t=1}^n$ with an energy measurement $|\mathbf{a}_i^\top \mathbf{x}|^2$ and aggregate the average energy measurement⁵:

$$y_i = \frac{1}{n} \sum_{t=1}^n \left| \mathbf{a}_i^\top \mathbf{x}_t \right|^2 + \eta_i = \left\langle \mathbf{a}_i \mathbf{a}_i^\top, \mathbf{S} \right\rangle + \eta_i$$

⁵Since only finite samples are available, we have $\mathbf{S} = \mathbf{\Sigma}^* + \mathbf{E}$ with $\mathbf{E}$ a bias term.

Related Work I

The *sparsity* assumption: a majority of the off-diagonal elements of Σ^* are zero or nearly so.

- Positive definite non-convex penalized covariance estimator (Quan 2023)

$$\hat{\Sigma} = \arg \min_{\Sigma \succ 0} \left\{ \frac{1}{2} \|\Sigma - S\|_F^2 - \tau \log \det \Sigma + \sum_{i \neq j} p_\lambda(|\Sigma_{ij}|) \right\}$$

- 😊 is *always* positive definite,
- 😊 is *easy* to obtain by iterative algorithm due to its convexity,
- 😞 need the full data samples.

Sampling Model

Consider n independent observations $\{\mathbf{x}_t\}_{t=1}^n$, each drawn from a zero-mean random vector $\mathbf{x}$ with covariance matrix $\boldsymbol{\Sigma}^*$. Given m sensing vectors $\{\mathbf{a}_i\}_{i=1}^m$, the quadratic measurement y_i , $i = 1, \dots, m$, is given by

$$\mathbf{y} = \mathbf{A}_{\otimes} \text{vec}(\mathbf{S}) + \boldsymbol{\eta} = \mathcal{A}(\mathbf{S}) + \boldsymbol{\eta},$$

where $\mathbf{y} := [y_1, \dots, y_m]^\top$, $\boldsymbol{\eta} := [\eta_1, \dots, \eta_m]^\top$ are additive measurement noises,

$\mathbf{A}_{\otimes} = \begin{bmatrix} (\mathbf{a}_1 \otimes \mathbf{a}_1) & \cdots & (\mathbf{a}_m \otimes \mathbf{a}_m) \end{bmatrix}^\top$ and $\text{vec}(\mathbf{S})$ denotes the vectorization of $\mathbf{S}$ obtained by stacking its columns, and $\mathcal{A} : \mathbb{R}^{d \times d} \mapsto \mathbb{R}^m$ is a linear operator.

Additional assumptions:

- The sensing vectors $\{\mathbf{a}_i\}_{i=1}^m$ are i.i.d. sub-Gaussian random variables with zero mean and identity covariance.
- The measurement noises $\{\eta_i\}_{i=1}^m$ are i.i.d. sub-exponential random variables with mean 0 and variance proxy σ^2 .

Problem Formulation

- We propose to estimate the sparse covariance matrices from quadratic measurements using the non-convex penalty

$$\min_{\Sigma \succ \mathbf{0}} \left\{ \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_{\text{F}}^2 - \tau \log \det \Sigma + \sum_{i,j} p_{\lambda}(|\Sigma_{ij}|) \right\}.$$

- Assumptions on the non-convex penalty function $p_{\lambda}(\cdot)$:
 - (i) $p_{\lambda}(t)$ is non-decreasing on $[0, +\infty)$ with $p_{\lambda}(0) = 0$ and is differentiable on $(0, +\infty)$;
 - (ii) $0 \leq p'_{\lambda}(t_1) \leq p'_{\lambda}(t_2) \leq \lambda$ for all $t_1 \geq t_2 \geq 0$ and $\lim_{t \rightarrow 0} p'_{\lambda}(t) = \lambda$;
 - (iii) There exists an $\alpha > 0$ such that $p'_{\lambda}(t) = 0$ for $t \geq \alpha\lambda$.

Prototypical examples of the non-convex penalty function $p_{\lambda}(\cdot)$: SCAD and MCP

Challenges

$$\min_{\Sigma \succ 0} \left\{ \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_F^2 - \tau \log \det \Sigma + \sum_{i,j} p_\lambda(|\Sigma_{ij}|) \right\}.$$

- It is a non-convex problem.
- If we directly apply an iterative algorithm, e.g., coordinate descent (Rothman 2012),
 - the global optimum may not be obtainable,
 - the local optimums are in general hard to be characterized.

Question: how to develop numerical algorithms with provable statistical guarantees?

Optimization Algorithm

Algorithm 1: Majorization-Minimization Based Multistage Convex Relaxation

Initialize $\tilde{\Sigma}^{(0)} = \mathbf{I}$

for $k = 1, 2, \dots, K$ **do**

 update $\Lambda_{ij}^{(k-1)} = p'_\lambda(|\tilde{\Sigma}_{ij}^{(k-1)}|)$;

 obtain $\tilde{\Sigma}^{(k)}$ by solving

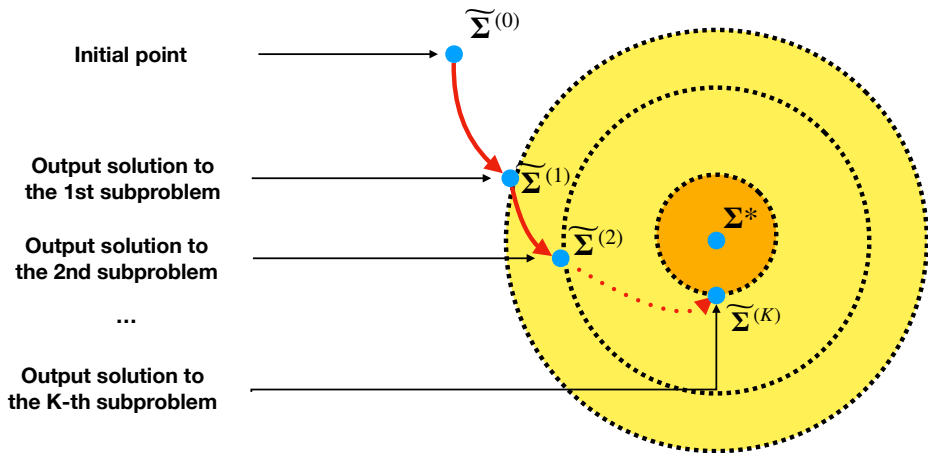
$$\min_{\Sigma \succ \mathbf{0}} \left\{ \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_{\text{F}}^2 - \tau \log \det \Sigma + \sum_{i,j} p_\lambda(|\Sigma_{ij}|) \right\},$$

$k = k + 1$;

end for.

- Due to numerical optimization error in practice, we can only compute an **approximate** solution (ε -optimal solution) $\tilde{\Sigma}$ instead of the **optimal** solution to each subproblem.

Algorithm Illustration



- Our algorithm can guarantee that an **approximate local optimum** enjoys the **optimal** statistical property.

Essential Assumptions

- The true covariance matrix Σ^* satisfies

$$0 < \frac{1}{\kappa} \leq \lambda_{\min}(\Sigma^*) \leq \lambda_{\max}(\Sigma^*) \leq \kappa < \infty,$$

for some constant $\kappa \geq 1$.

- There exist universal constants α and μ such that

$$\|\Sigma_{\mathcal{S}}^*\|_{\min} = \min_{(i,j) \in \mathcal{S}} |\Sigma_{ij}^*| \geq (\alpha + \mu) \lambda,$$

where $\mu \in (0, \alpha)$ satisfies $p'_\lambda(\mu\lambda) \geq \frac{\lambda}{2}$.

- Restricted Strong Convexity (RSC) & Restricted Strong Smoothness (RSS):** For the function f , there exists some $\infty > \rho^+ > \rho^- > 0$ such that, for all $\Delta \in \mathcal{B}\left(\Sigma^*, \frac{\rho^-}{4\tau\kappa}\right)$,

$$f(\Sigma + \Delta) \geq f(\Sigma) + \langle \nabla f(\Sigma), \Delta \rangle + \frac{\rho^-}{2} \|\Delta\|_F, \quad (1)$$

$$f(\Sigma + \Delta) \leq f(\Sigma) + \langle \nabla f(\Sigma), \Delta \rangle + \frac{\rho^+}{2} \|\Delta\|_F. \quad (2)$$

Theoretical Results I

Let $\mathcal{S} = \{(i, j) \mid \Sigma_{ij}^* \neq 0\}$ and $|\mathcal{S}| = s$. Define $f(\Sigma) = \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_F^2 - \tau \log \det \Sigma$.

Theorem 1 (contraction property)

Under some standard assumptions, with probability exceeding $1 - c_1 \exp(-c_2 m)$ for some $c_1, c_2 > 0$, then the ε -optimal solution $\tilde{\Sigma}^{(k)}$ ($1 \leq k \leq K$) satisfies the following contraction property:

$$\left\| \tilde{\Sigma}^{(k)} - \Sigma^* \right\|_F \leq \frac{1}{\rho} \left(\underbrace{\|(\nabla f(\Sigma^*))_{\mathcal{S}}\|_F}_{\text{oracle rate}} + \underbrace{\varepsilon \sqrt{s}}_{\text{optimization error}} \right) + \underbrace{\delta \left\| \tilde{\Sigma}^{(k-1)} - \Sigma^* \right\|_F}_{\text{contraction}},$$

where $\delta \in (0, 1)$ is the contraction parameter, provided that $m = \mathcal{O}((s + s^\diamond) \log^2(d/(s + s^\diamond)))$ with $s^\diamond \geq \beta s$ for some universal constant β .

Theoretical Results II

Corollary 2 (oracle rate)

Under some standard assumptions, let $\mathbf{x}$ be a sub-Gaussian random vector with mean zero and covariance Σ^* and $\{\mathbf{x}_i\}_{i=1}^n$ be a collection of i.i.d. samples drawn from $\mathbf{x}$, if $\lambda \asymp \sqrt{\frac{\log d}{mn}}$,

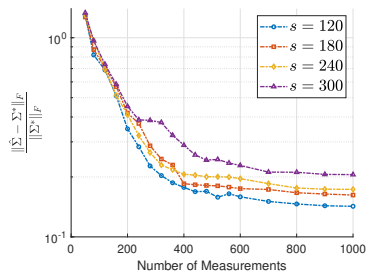
$\tau \lesssim \sqrt{\frac{1}{mn}} \left\| (\Sigma^*)^{-1} \right\|_{\max}^{-1}$, $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$, and $K \gtrsim \log(\lambda \sqrt{mn}) \gtrsim \log \log d$, then the ε -optimal solution $\tilde{\Sigma}^{(K)}$ satisfies

$$\left\| \tilde{\Sigma}^{(K)} - \Sigma^* \right\|_F \lesssim \sqrt{\frac{s}{mn}}$$

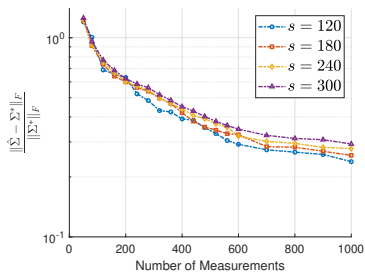
with high probability.

- The *oracle rate* refers to the statistical convergence rate of the *oracle estimator*, defined as $\hat{\Sigma}^O = \arg \min_{\Sigma: \Sigma_{\bar{S}}=0} f(\Sigma)$, which knows the true coefficients in advance. By the mean value theorem, it is easy to obtain that $\left\| \hat{\Sigma}^O - \Sigma^* \right\|_F \lesssim \left\| (\nabla f(\Sigma^*))_{\mathcal{S}} \right\|_F \lesssim \sqrt{\frac{s}{mn}}$.

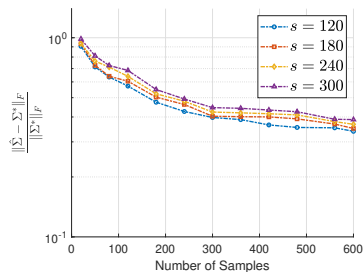
Numerical Experiments I



(d)



(e)



(f)

Figure 2: The FRE of the estimated covariance matrices is examined in three distinct scenarios: (a) the true covariance without added noise; (b) the sample covariance with a noise parameter of $\gamma = 0.1$ and $n = 50$; (c) the sample covariance with a noise parameter of $\gamma = 0.1$ and $m = 300$;

Numerical Experiments II

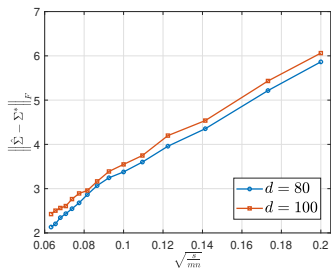


Figure 3: The oracle rate of “sprandsym” Matrix with $s = 120$ and $\gamma = 0.1$.

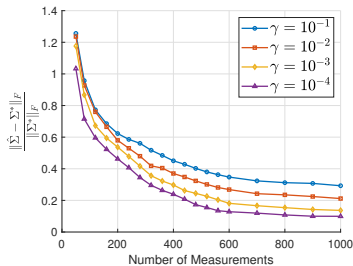


Figure 4: The FRE of the estimated covariance matrices for different noise levels when $s = 300$.

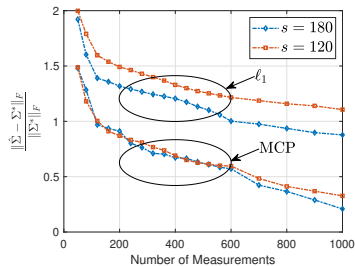


Figure 5: The FRE of the estimated covariance matrices for different sparsity levels with noise level $\gamma = 10^{-1}$ (ℓ_1 v.s. MCP).

Numerical Experiments III

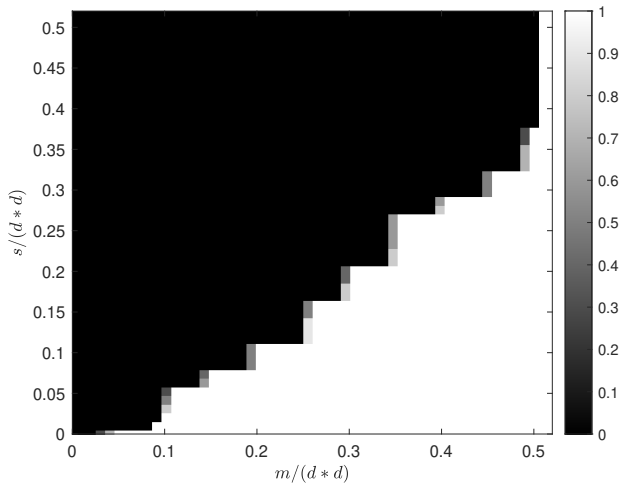


Figure 6: The rate of successful covariance reconstruction when $d = 100$.

Conclusion

- We have proposed a novel approach for large sparse covariance matrix estimation from quadratic measurements using the non-convex penalty and presented both the theoretical and empirical results.
- To the best of our knowledge, this is **the first work** to obtain the **optimal statistical rate** for large sparse covariance matrix estimation from quadratic measurements.

Thank you!